RETAIL PURCHASE PATTERN ANALYSIS

Abstract:

This project focuses on analyzing retail purchase patterns across both online and offline stores using data warehousing and data mining techniques. The study aims to understand customer behavior and identify meaningful patterns in their buying habits. For this purpose, two separate datasets representing online and offline retail scenarios were considered. In the online retail analysis, customer purchase data was examined to identify product preferences and trends. Based on these insights, an advertisement recommendation system was designed to suggest relevant products to users according to the categories they frequently purchase.

On the other hand, the offline store analysis focuses on understanding customer loyalty and purchasing behavior within physical stores. Using this information, a system was proposed to provide personalized coupons and discounts to customers. These offers are based on the types of products they buy and their level of loyalty towards the store. Overall, the project demonstrates how data-driven techniques can be effectively used to enhance customer engagement and improve sales strategies in both online and offline retail environments.

Information about Datasets:

For this project, two different datasets were utilized to perform both online and offline retail analysis. The first dataset, known as the Online Retail dataset, was used for analyzing online shopping patterns and contains approximately 50,000 records. The second dataset, referred to as the Clothes dataset, was used for offline store analysis and includes around 3,500 entries. Both datasets were sourced from Kaggle, ensuring reliability and relevance for the study. The attribute tables for each dataset are presented below for further reference.

Attribute table for “OnlineRetail.csv”.

Attribute Name	Description
InvoiceNo	Unique ID for each transaction
StockCode	Product ID
Description	Name of the product
Quantity	Number of items purchased
InvoiceDate	Date and time of transaction
UnitPrice	Price per unit
CustomerID	Unique customer identifier
Country	Country of the customer

Attribute table for “shopping_trends.csv”

Attribute Name	Description
Customer ID	Unique identifier assigned to each customer
Age	Age of the customer

Gender	Gender of the customer
Item Purchased	Name of the clothing item purchased
Category	Type of product (e.g., clothing category)
Purchase Amount (USD)	Total amount spent on the purchase
Location	Store location or region of purchase
Size	Size of the purchased item (e.g., S, M, L)
Color	Color of the product
Season	Season during which the purchase was made
Review Rating	Customer rating for the purchased item
Subscription Status	Indicates whether the customer is subscribed to services/offers
Payment Method	Method used for payment (e.g., card, cash)
Shipping Type	Type of shipping selected
Discount Applied	Indicates whether a discount was applied
Promo Code Used	Indicates whether a promotional code was used
Previous Purchases	Number of previous purchases made by the customer
Preferred Payment Method	Customer's preferred mode of payment
Frequency of Purchases	How often the customer makes purchases

Introduction:

In today's retail industry, a large amount of data is generated from both online and offline transactions, but much of this data is not effectively utilized for decision-making. The main aim of this project, *Retail Purchase Pattern Analysis*, is to analyze customer purchasing behavior and extract meaningful insights using data warehousing and data mining techniques. This project consists of two case studies, focusing on online and offline retail datasets, respectively, to understand different types of customer interactions. In the online retail case, association rule mining is used to identify product relationships and generate recommendations. In the offline clothing dataset, a Random Forest model is applied to determine important features influencing product categorization and customer preferences. Based on these insights, personalized recommendations and coupon strategies are designed. Overall, this project highlights how data-driven approaches can help improve customer experience and business performance in the retail sector.

Problem Statement:

The rapid growth of retail businesses has led to the accumulation of large volumes of transactional data, but many organizations lack efficient systems to organize and utilize this data effectively. In many cases, data from online and offline stores is handled separately without any structured analysis, leading to missed opportunities in understanding customer behavior. Additionally, marketing strategies are often generalized rather than personalized, which reduces their effectiveness in influencing customer decisions. Another issue is the lack of integration between customer purchase history and promotional strategies, making it difficult to provide relevant recommendations or rewards.

This project addresses these challenges by focusing on how retail data can be systematically stored, processed, and analyzed to support better business decisions. It aims to overcome the limitations of unstructured data usage by applying appropriate analytical techniques to derive useful insights. The goal is to create a more targeted approach for customer engagement by linking purchasing patterns with suitable recommendation and reward mechanisms in both online and offline retail contexts.

Our Approach:

This project follows a dual case study approach to analyze retail purchase patterns across both online and offline environments. The use of two separate case studies allows for a more accurate understanding of customer behavior, as each dataset differs in terms of structure, context, and business objectives. By applying appropriate data mining and machine learning techniques to each dataset, the project aims to extract meaningful insights and design effective recommendation strategies.

Case Study 1: Online Retail Dataset

The first case study focuses on an online retail dataset that captures transactional data generated through e-commerce platforms. This dataset reflects customer interactions in a digital environment, where purchasing behavior is influenced by browsing patterns and product visibility.

In this case study, market basket analysis is performed using the Apriori algorithm to identify frequent itemsets and discover associations between products. The analysis helps in understanding which products are commonly purchased together and enables the generation of recommendation rules. These insights are further used to infer customer preferences and improve product recommendation strategies.

Due to its detailed and time-stamped nature, the online dataset is well-suited for pattern mining and recommendation system development.

Case Study 2: Offline Retail (Clothing) Dataset

The second case study is based on an offline retail dataset related to clothing products. Unlike the online dataset, this data includes structured information about product attributes and customer types, reflecting in-store purchasing behavior.

This dataset is analyzed using a Random Forest classification model to identify the most important features influencing product categorization. Factors such as fabric, style, and price range, along with customer segmentation attributes, are considered in the analysis. Based on the model's output, insights are derived to understand customer preferences and to design coupon recommendation strategies tailored to different customer types.

This approach helps in improving targeted marketing and enhancing customer engagement in offline retail settings.

Need for Two Separate Case Studies

The decision to use two separate case studies in this project is mainly due to the fundamental differences between online and offline retail data, which make it difficult to combine them into a single unified analysis. Each dataset represents a different retail environment with distinct patterns of customer behavior, data characteristics, and analytical objectives. Treating them as separate case studies allows for more accurate analysis and the application of appropriate techniques suited to each context.

One of the key differences lies in customer behavior. In online retail, purchasing decisions are largely influenced by convenience, browsing history, and system-generated recommendations, whereas in offline retail, decisions are driven more by physical interaction with products, immediate requirements, and the in-store experience. Along with this, the structure of the datasets also varies significantly. Online retail data mainly consists of transactional records such as product IDs, purchase combinations, and timestamps, while offline retail data includes detailed product attributes and customer-related features, making it more suitable for classification-based analysis.

Furthermore, the objectives of the analysis in both cases are different. The online dataset is primarily used for association rule mining and recommendation generation, while the offline dataset focuses on identifying important features and understanding customer segments using classification techniques. Differences in customer demographics and purchasing intent also play a role, as online platforms cater to a wider and more diverse audience, whereas offline retail is often location-specific with more clearly defined customer groups. Therefore, analyzing these datasets separately ensures better insight generation and more effective results.

Implementation Details:

Case Study 1: Online Retail Dataset

Data Collection and Preprocessing:

The dataset used in this study consists of transactional records from an online retail store. Each record contains details such as invoice number, product description, quantity, customer ID, and transaction date.

The first step involved data cleaning and preprocessing to ensure the quality and reliability of the analysis. Records with missing customer IDs were removed, as customer-level analysis requires valid identifiers. Transactions with negative or zero quantities were also eliminated, since they represent returns or invalid purchases and do not reflect actual buying behavior.

Additionally, the InvoiceDate field was converted into a datetime format to enable time-based computations, and the CustomerID field was treated as a categorical variable rather than a numerical one.

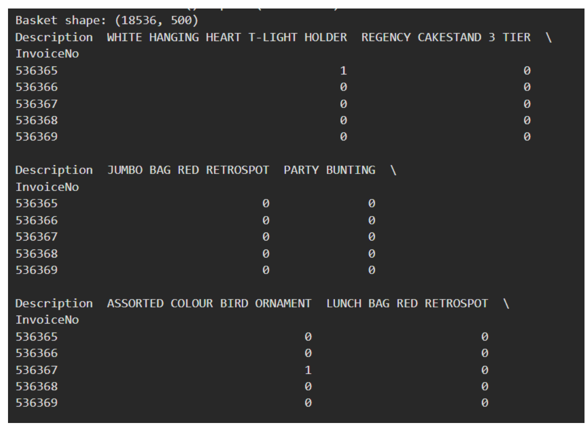
Basket Matrix Construction:

To perform market basket analysis, the transactional data was transformed into a basket matrix representation. In this structure, each row corresponds to a unique transaction (invoice), and each column represents a product. The values in the matrix indicate whether a product was purchased in a given transaction.

This transformation was achieved using a pivot operation, which aggregates quantities of products per invoice. The matrix was further converted into a binary format, where a value of 1 indicates the presence of a product in a transaction, and 0 indicates its absence.

Although all products are initially considered, only the top 500 most frequently purchased products were retained to reduce sparsity and computational complexity.

This representation is essential because the Apriori algorithm operates on transactional data in terms of item presence rather than quantity.



```
Basket shape: (18536, 500)
Description  WHITE HANGING HEART T-LIGHT HOLDER  REGENCY CAKESTAND 3 TIER  \
InvoiceNo
536365                                        1                        0
536366                                        0                        0
536367                                        0                        0
536368                                        0                        0
536369                                        0                        0

Description  JUMBO BAG RED RETROSPOT  PARTY BUNTING  \
InvoiceNo
536365                             0              0
536366                             0              0
536367                             0              0
536368                             0              0
536369                             0              0

Description  ASSORTED COLOUR BIRD ORNAMENT  LUNCH BAG RED RETROSPOT  \
InvoiceNo
536365                                   0                        0
536366                                   0                        0
536367                                   1                        0
536368                                   0                        0
536369                                   0                        0
```

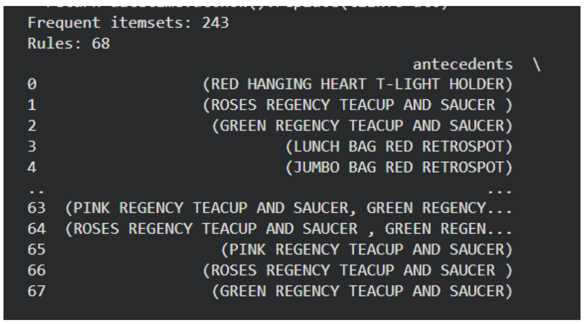
Binary basket matrix showing product presence across transactions

Market Basket Analysis Using Apriori Algorithm:

The Apriori algorithm was applied to identify frequent itemsets, which are combinations of products that appear together in transactions above a specified support threshold. A minimum support value of 0.02 was used to balance computational efficiency and meaningful pattern discovery.

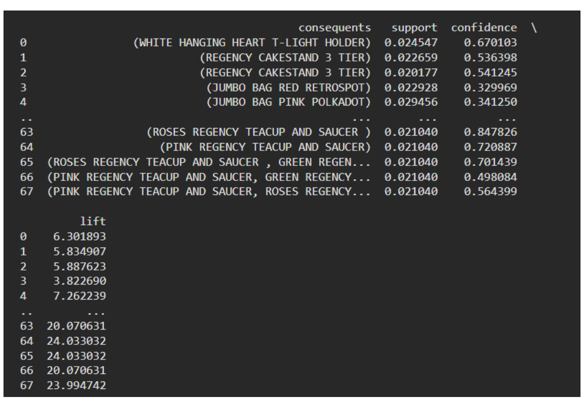
Apriori works by generating candidate itemsets and pruning those that do not satisfy the minimum support condition, based on the principle that all subsets of a frequent itemset must also be frequent.

From the frequent itemsets, association rules were generated using metrics such as support, confidence, and lift. These rules help identify relationships between products, such as which items are likely to be purchased together.



```
Frequent itemsets: 243
Rules: 68
                                         antecedents  \
0                     (RED HANGING HEART T-LIGHT HOLDER)
1                     (ROSES REGENCY TEACUP AND SAUCER )
2                      (GREEN REGENCY TEACUP AND SAUCER)
3                              (LUNCH BAG RED RETROSPOT)
4                              (JUMBO BAG RED RETROSPOT)
..                                                   ...
63  (PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY...
64  (ROSES REGENCY TEACUP AND SAUCER , GREEN REGEN...
65                      (PINK REGENCY TEACUP AND SAUCER)
66                    (ROSES REGENCY TEACUP AND SAUCER )
67                     (GREEN REGENCY TEACUP AND SAUCER)
```

Sample of frequent itemsets identified using Apriori



```
                                         consequents   support  confidence  \
0                 (WHITE HANGING HEART T-LIGHT HOLDER)  0.024547    0.670103
1                           (REGENCY CAKESTAND 3 TIER)  0.022659    0.536398
2                           (REGENCY CAKESTAND 3 TIER)  0.020177    0.541245
3                            (JUMBO BAG RED RETROSPOT)  0.022928    0.329969
4                            (JUMBO BAG PINK POLKADOT)  0.029456    0.341250
..                                                 ...       ...         ...
63                    (ROSES REGENCY TEACUP AND SAUCER )  0.021040    0.847826
64                      (PINK REGENCY TEACUP AND SAUCER)  0.021040    0.720887
65  (ROSES REGENCY TEACUP AND SAUCER , GREEN REGEN...  0.021040    0.701439
66  (PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY...  0.021040    0.498084
67  (PINK REGENCY TEACUP AND SAUCER, ROSES REGENCY...  0.021040    0.564399

         lift
0    6.301893
1    5.834907
2    5.887623
3    3.822690
4    7.262239
..        ...
63  20.070631
64  24.033032
65  24.033032
66  20.070631
67  23.994742
```

Association rules showing relationships between products

<u>Recommendation and Advertisement System:</u>

Based on the generated association rules, a recommendation system was developed. Given a selected product, the system identifies rules where the product appears either as an antecedent or a consequent and recommends associated items ranked by confidence.

This approach enables the system to suggest products that are frequently purchased together, thereby improving cross-selling opportunities and enhancing customer experience.

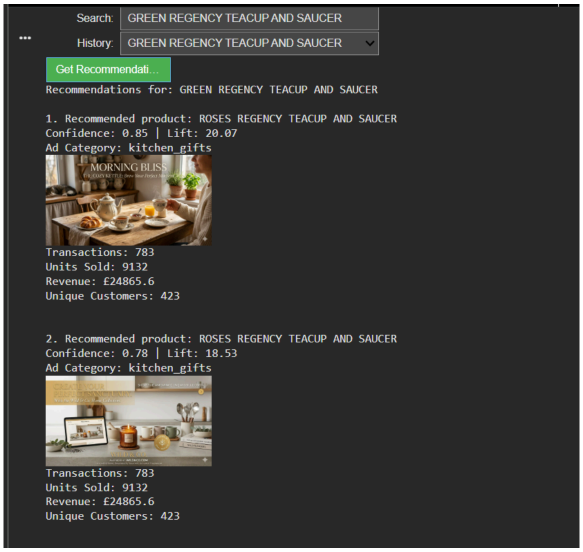
In addition to product recommendations, an advertisement recommendation system was also implemented to enhance user engagement. Products were categorized into six broad categories based on their textual descriptions:

- Kitchen
- Home Decor
- Gifts
- Toys
- Seasonal
- Stationery

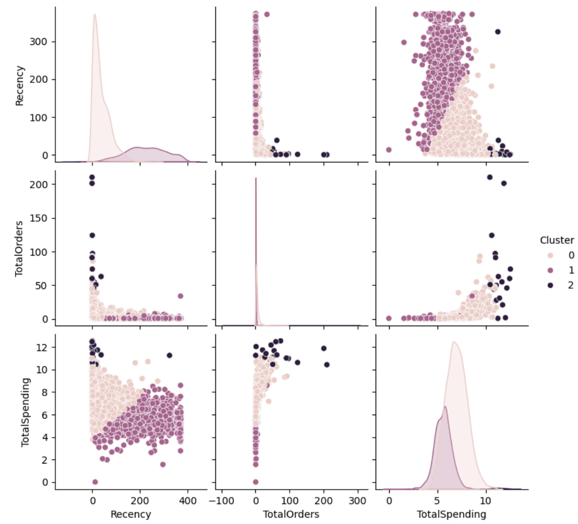
This categorization was performed using string matching techniques, where keywords in the product name were used to map each item to a relevant category.

Once a category is identified for a selected product, the system retrieves a corresponding advertisement image from a predefined image dataset associated with that category. These images are displayed alongside recommendations to simulate a real-world e-commerce interface.

This hybrid approach combines data-driven recommendations (Apriori) with content-based advertisement selection, thereby improving both relevance and visual engagement.



Product recommendations generated using association rules and a category-based advertisement system using product classification



Cluster	Recency	TotalOrders	TotalSpending
0	38.528039	5.015883	6.976274
1	224.758900	1.466828	5.563372
2	24.277778	69.388889	11.390129

Computed RFM features for customer segmentation

Customer Segmentation using RFM Analysis:

To understand customer behavior, RFM (Recency, Frequency, Monetary) analysis was performed. For each customer:

- Recency measures how recently a customer made a purchase
- Frequency measures how often purchases are made
- Monetary measures total spending

Additional features, such as the total number of items purchased and average basket size, were also computed to capture more detailed behavioral patterns.

Since monetary values are often skewed, a logarithmic transformation was applied to normalize the distribution. All features were then standardized to ensure equal contribution during clustering.

Clustering Using K-Means Algorithm:

K-Means clustering was applied to segment customers into groups based on their purchasing behavior. The number of clusters was set to three, representing:

- Occasional customers
- Regular customers
- High-value (VIP) customers

K-Means was chosen due to its simplicity, efficiency, and suitability for numerical data. Feature scaling ensured that no single feature dominated the clustering process.

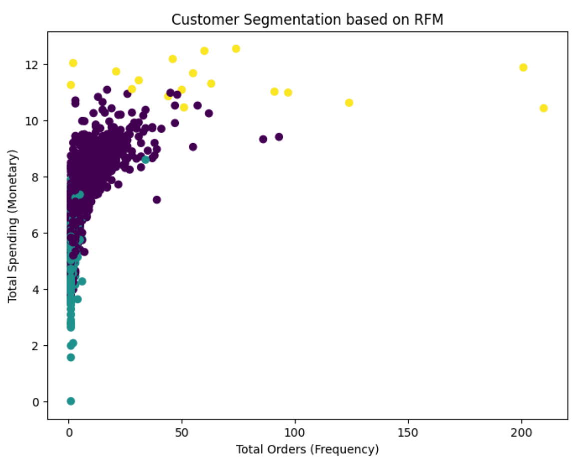
```
Cluster
0    3085
1    1236
2      18
Name: count, dtype: int64
```

Distribution of customers across different clusters

Visualization and Analysis:

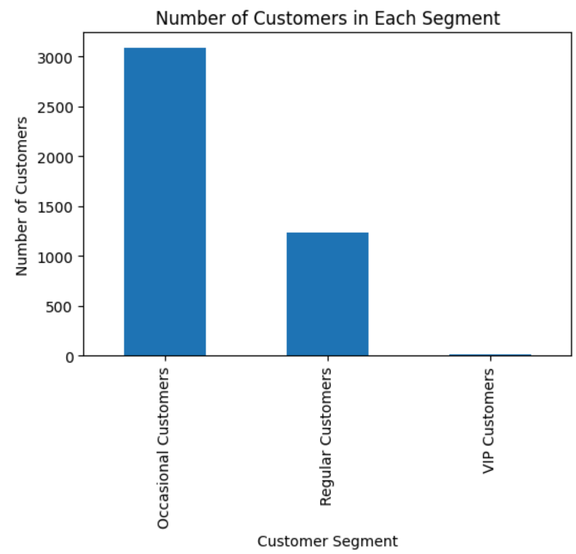
A scatter plot was used to visualize the relationship between purchase frequency and total spending, with different colors representing customer clusters.

This plot helps identify high-value customers (high frequency and spending) and low-engagement customers.



Customer segmentation based on frequency and spending

A bar chart was used to show the number of customers in each cluster. This helps understand the distribution of customer types and identify which segment dominates.



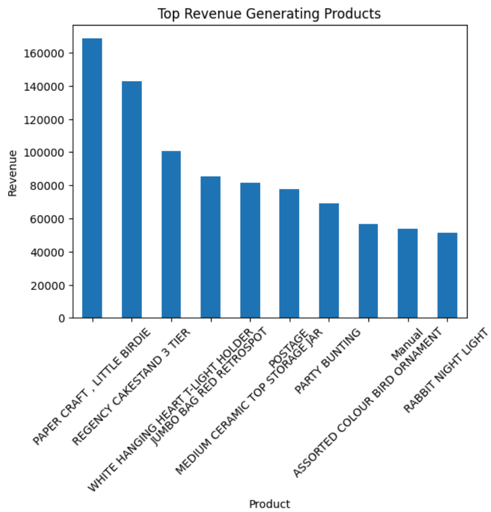
Distribution of customers across segments

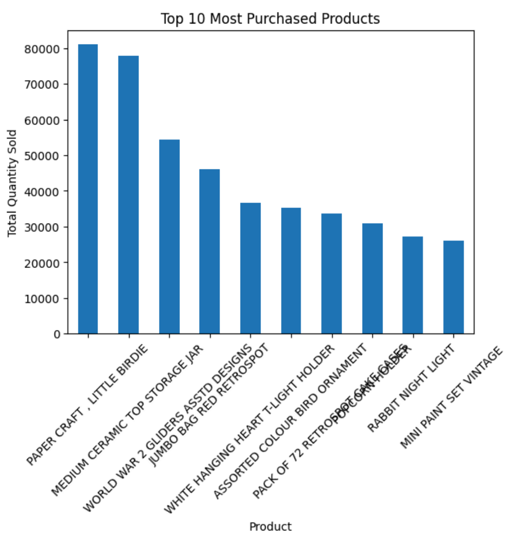
Results and Observations:

The analysis of product purchase frequency shows that certain items dominate transactions, indicating high customer demand and potential candidates for promotion.

```
Top Selling Products
Description
WHITE HANGING HEART T-LIGHT HOLDER    2028
REGENCY CAKESTAND 3 TIER              1724
JUMBO BAG RED RETROSPOT               1618
ASSORTED COLOUR BIRD ORNAMENT         1408
PARTY BUNTING                         1397
LUNCH BAG RED RETROSPOT               1316
SET OF 3 CAKE TINS PANTRY DESIGN      1159
LUNCH BAG  BLACK SKULL.               1105
POSTAGE                               1099
PACK OF 72 RETROSPOT CAKE CASES       1068
Name: count, dtype: int64
```

Revenue analysis reveals that high-selling products are not always the highest revenue generators, highlighting the importance of pricing strategy.

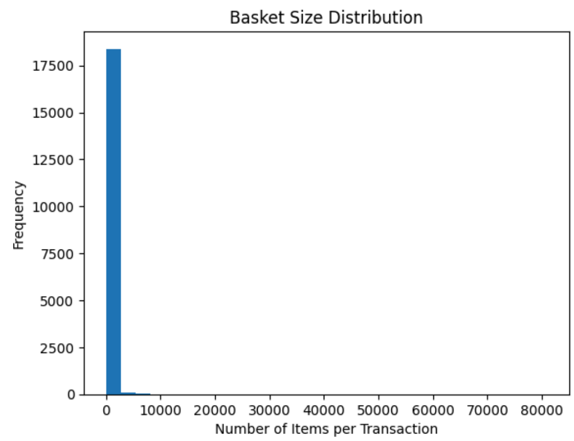




Analysis of product preferences across clusters shows that different customer segments exhibit distinct purchasing patterns, enabling targeted marketing strategies.

```
Cluster  Cluster  Description
0        0        WORLD WAR 2 GLIDERS ASSTD DESIGNS     47970
                  JUMBO BAG RED RETROSPOT               35721
                  ASSORTED COLOUR BIRD ORNAMENT         31341
                  POPCORN HOLDER                        28447
                  WHITE HANGING HEART T-LIGHT HOLDER    27520
1        1        ASSTD DESIGN 3D PAPER STICKERS        12786
                  WORLD WAR 2 GLIDERS ASSTD DESIGNS      5197
                  ASSORTED COLOURS SILK FAN              2876
                  WHITE HANGING HEART T-LIGHT HOLDER     2338
                  GIRLS ALPHABET IRON ON PATCHES         2304
2        2        PAPER CRAFT , LITTLE BIRDIE           80995
                  MEDIUM CERAMIC TOP STORAGE JAR        75784
                  BROCADE RING PURSE                    10562
                  JUMBO BAG RED RETROSPOT                9403
                  PACK OF 72 RETROSPOT CAKE CASES        8693
Name: Quantity, dtype: int64
```

The distribution of basket size indicates the average number of items per transaction, providing insights into customer buying behavior.



<u>Conclusion:</u>

This study demonstrates how transactional data can be effectively utilized to extract meaningful insights and improve retail decision-making. The application of the Apriori algorithm enabled the identification of strong product associations, forming the basis for a recommendation system that enhances cross-selling opportunities. Additionally, RFM-based customer segmentation using KMeans clustering provided a clear understanding of customer behavior, distinguishing between occasional, regular, and high-value customers. Further analysis of product demand, revenue contribution, and cluster-wise preferences offered valuable business insights for targeted marketing strategies. The integration of a category-based advertisement system added a practical dimension by improving user engagement through relevant visual recommendations. Overall, the project highlights how combining data mining techniques with simple modeling approaches can significantly enhance personalization and operational efficiency in retail environments.

Case Study 2: Offline Retail (Clothing) Dataset

Data Preprocessing:

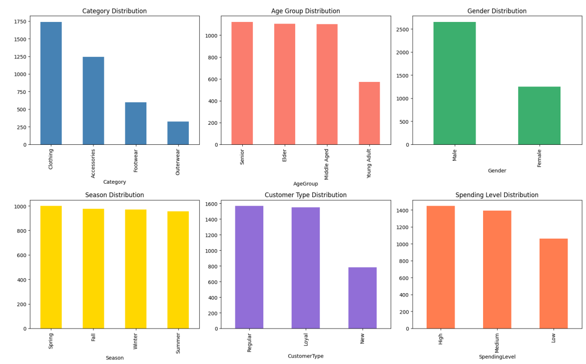
The dataset was preprocessed using the following steps:

1. Handling categorical variables using encoding techniques
2. Creating bins for continuous variables (Age, Purchase Amount)
3. Verifying absence of null values
4. Structuring data for visualization and modeling

This step ensures that the data is clean, structured, and suitable for mining and machine learning tasks.

Exploratory Data Analysis(EDA):

To understand the overall distribution and characteristics of the dataset, exploratory data analysis was performed using multiple visualizations. These plots provide a high-level overview of how different attributes such as category, age group, gender, season, customer type, and spending level, are distributed across the dataset. This step is crucial as it helps in identifying dominant trends and potential imbalances before performing a deeper analysis.

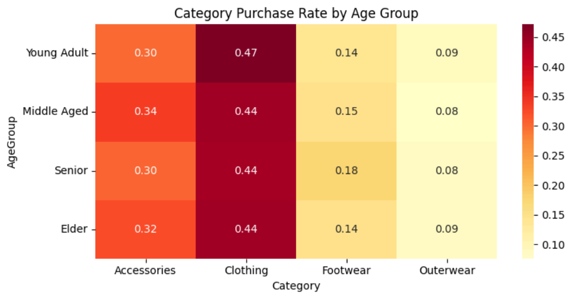


Distribution of key attributes, including category, age group, gender, season, customer type, and spending level.

The analysis reveals that the Clothing category dominates overall purchases, indicating it is the most popular segment. Customer distribution across age groups is relatively balanced, ensuring that the dataset represents diverse demographics. Male customers contribute slightly more to total purchases compared to female customers. Seasonal distribution appears consistent, suggesting stable demand throughout the year. Additionally, a significant portion of customers falls under Regular and Loyal categories, indicating repeated engagement, while medium and high spending groups dominate purchase behavior.

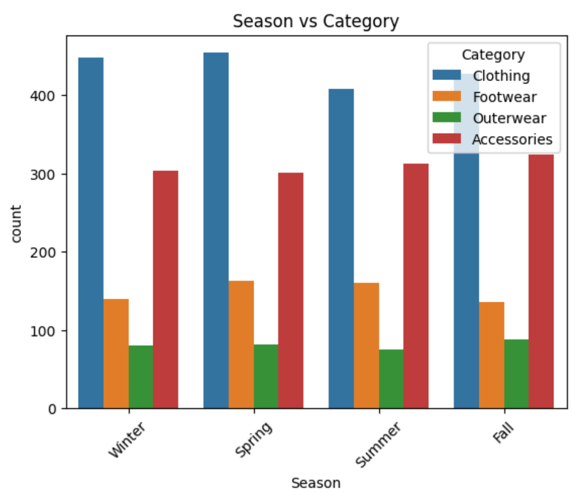
Pattern Analysis:

To uncover deeper relationships within the data, cross-dimensional analysis was performed by comparing multiple attributes such as age group, season, gender, and customer type with product categories. This helps in identifying how different customer segments influence purchasing decisions.



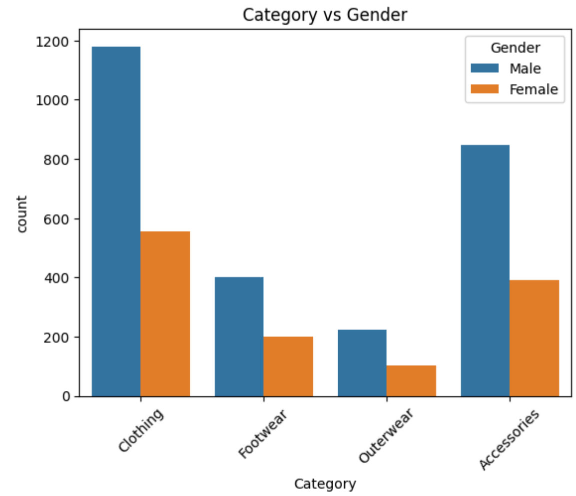
Category purchase distribution across different age groups.

The heatmap illustrates that Clothing remains the most preferred category across all age groups, followed by Accessories. Footwear and Outerwear show relatively lower purchase frequencies. This indicates that customer preferences are consistent across age segments, with minimal variation, suggesting that certain product categories have universal appeal.



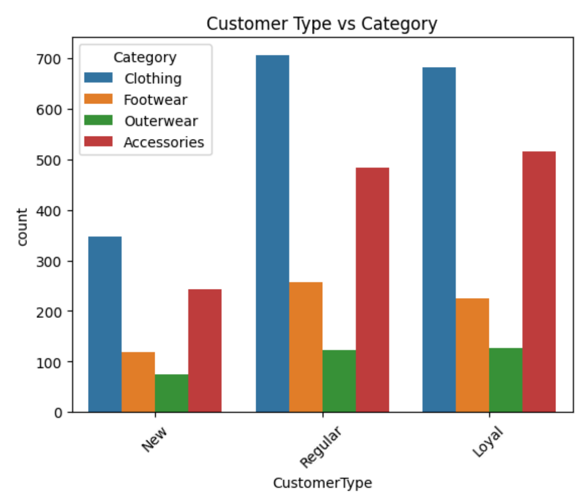
Variation of product category purchases across different seasons.

The seasonal analysis highlights that purchasing behavior varies with seasons. While Clothing remains consistently high in demand, Accessories show higher preference during Spring, and Outerwear sees increased demand during colder seasons. This demonstrates that seasonal factors play an important role in influencing purchasing patterns.



Comparison of product category purchases across genders.

The gender-based analysis shows that male customers contribute a higher number of purchases overall. However, both genders exhibit similar category preferences, with Clothing being dominant. This suggests that while purchase volume differs, product preferences remain largely consistent across genders.

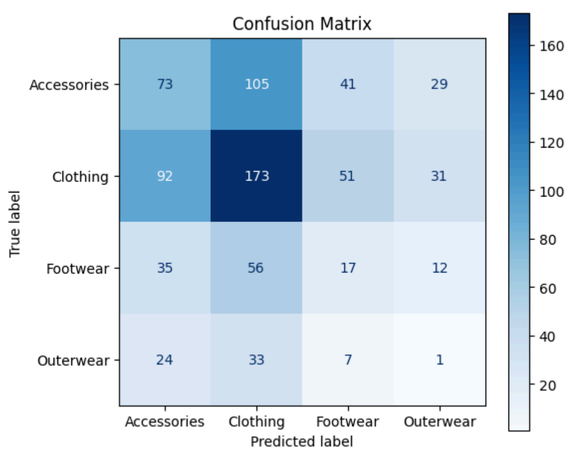


Product category preferences across different customer types.

The analysis of customer types reveals that Loyal customers make the highest number of purchases and show a strong preference for Clothing. In contrast, new customers display more diverse purchasing behavior across categories. This indicates that customer loyalty is associated with consistent buying patterns.

Machine Learning Model:

To extend the analysis into predictive capabilities, a Random Forest Classifier was implemented to predict the product category based on customer attributes. The model uses features such as age group, gender, spending level, customer type, season, subscription status, discount usage, and purchase frequency. The dataset was split into training and testing sets using an 80:20 ratio with stratification to preserve class distribution. Random Forest was chosen due to its robustness and ability to handle categorical and non-linear relationships effectively.

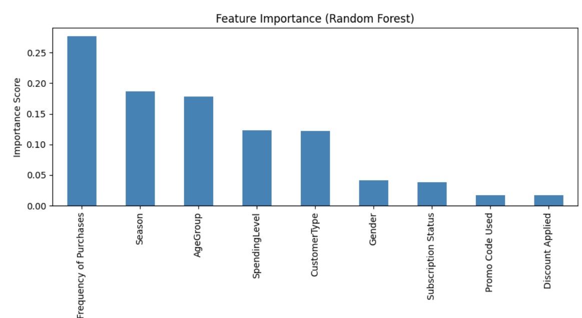


Confusion matrix showing classification performance of the Random Forest model.

The model achieved an accuracy of approximately 83–84%, indicating good predictive performance. The confusion matrix shows that the Clothing category is predicted with high accuracy, while some misclassifications occur between similar categories such as Accessories and Clothing or Footwear and Clothing. This suggests overlapping patterns in customer behavior across these categories.

Feature Importance Analysis:

To understand which factors influence predictions the most, feature importance analysis was performed using the trained Random Forest model. This helps in identifying the key drivers behind customer purchase behavior.



Importance of different features in predicting product categories.

The analysis shows that Frequency of Purchases is the most important feature, followed by Season, Age Group, and Spending Level. Features such as discount usage and promo codes have relatively lower importance. This indicates that intrinsic customer behavior and seasonal trends have a stronger impact on purchasing decisions than promotional factors.

<u>Recommendation System:</u>

Based on the insights obtained from data analysis and modeling, a rule-based recommendation system was developed to provide personalized suggestions and offers. The system recommends product categories using customer attributes such as age group, season, and spending level. It also generates discount coupons based on customer type and purchase frequency. Additionally, personalized messages are created to enhance user engagement.

The recommendation system integrates analytical insights with business logic to provide actionable outputs. For example, a high-spending young adult in winter is recommended Clothing products along with premium offers, while a loyal customer with frequent purchases receives higher discount incentives. This demonstrates how data mining results can be applied to real-world marketing strategies.

```
============================================================
PERSONALISED OFFER DEMO
============================================================

Profile: ('Young Adult', 'Winter', 'High', 'Loyal', 'Weekly')
✨ Upgrade your style! Premium collection on Winter-ready outfits tailored for Young Adults this Winter!
🎟  Use code LOYAL20 for 25% off on Clothing!
------------------------------------------------------------

Profile: ('Middle Aged', 'Spring', 'Medium', 'Regular', 'Monthly')
👌 Smart and practical picks! Great value picks on comfortable and stylish footwear tailored for Middle Ageds this Spring!
🎟  Use code REG10 for 13% off on Footwear!
------------------------------------------------------------

Profile: ('Senior', 'Fall', 'Low', 'New', 'Annually')
🧓 Comfort comes first! Budget-friendly deals on Fall jackets and hoodies tailored for Seniors this Fall!
🎟  Use code WELCOME5 for 5% off on Outerwear!
------------------------------------------------------------

Profile: ('Elder', 'Summer', 'Medium', 'Loyal', 'Every 3 Months')
🌿 Quality you can trust! Great value picks on watches, bags & stylish add-ons tailored for Elders this Summer!
🎟  Use code LOYAL20 for 22% off on Accessories!
------------------------------------------------------------
```

Sample of enriched DataFrame:

	AgeGroup	Season	SpendingLevel	CustomerType	Frequency of Purchases	RecommendedCategory	CouponOffer
0	Senior	Winter	Medium	Regular	Fortnightly	Footwear	Use code REG10 for 14% off on Footwear!
1	Young Adult	Winter	Medium	New	Fortnightly	Clothing	Use code WELCOME5 for 9% off on Clothing!
2	Senior	Spring	High	Regular	Weekly	Footwear	Use code REG10 for 15% off on Footwear!
3	Young Adult	Spring	High	Loyal	Weekly	Clothing	Use code LOYAL20 for 25% off on Clothing!
4	Senior	Spring	Medium	Loyal	Annually	Footwear	Use code LOYAL20 for 20% off on Footwear!
5	Senior	Summer	Low	Regular	Weekly	Footwear	Use code REG10 for 15% off on Footwear!
6	Elder	Fall	High	Loyal	Quarterly	Accessories	Use code LOYAL20 for 22% off on Accessories!
7	Middle Aged	Winter	Low	Regular	Weekly	Clothing	Use code REG10 for 15% off on Clothing!
8	Middle Aged	Summer	High	New	Annually	Accessories	Use code WELCOME5 for 5% off on Accessories!
9	Elder	Spring	Low	New	Quarterly	Outerwear	Use code WELCOME5 for 7% off on Outerwear!

<u>Conclusion:</u>

This case study demonstrates a complete data mining pipeline starting from data preprocessing and exploratory analysis to predictive modeling and recommendation generation. The analysis reveals key insights such as category dominance, seasonal influence, and the impact of customer segmentation on purchasing behavior. The machine learning model provides reliable predictions, while the recommendation system translates these insights into practical business applications. Overall, the study highlights the effectiveness of combining data mining techniques with machine learning to support intelligent decision-making in retail systems.

Overall Result from both the case studies:

The two case studies together provide a comprehensive understanding of retail purchase behavior across both online and offline settings. In the online retail analysis, transactional data enabled the discovery of strong product associations using Apriori, effective customer segmentation through RFM and KMeans, and insights into revenue patterns and basket composition. In contrast, the offline clothing dataset focused more on customer attributes and behavioral patterns, highlighting the influence of factors such as seasonality, spending levels, and purchase frequency. While the online approach emphasizes item-to-item relationships and market basket analysis, the offline approach complements it by focusing on customer-centric prediction and personalization. Both case studies demonstrate the importance of recommendation systems and targeted marketing strategies, such as product suggestions, coupon generation, and advertisement matching. Together, they show how data mining and machine learning techniques can enhance customer engagement, optimize sales, and support informed business decisions. Overall, this project highlights the significance of data-driven approaches in building intelligent and efficient retail systems.

Shortcomings of the corrent Solution:

Although both case studies successfully demonstrate the use of data mining techniques in retail analysis, the current implementation has certain limitations. In Case Study 1, the recommendation system is mainly based on support, confidence, and lift values from association rules, which may not always capture changing customer preferences or seasonal buying behavior. The advertisement recommendation system uses simple product categorization and string matching, which is basic and may not provide highly personalized recommendations. The KMeans clustering is performed using limited RFM features, and customer behavior beyond purchase frequency and spending is not deeply considered. In Case Study 2, the Random Forest model predicts product categories with good accuracy, but some categories such as Footwear and Outerwear still show misclassification due to overlapping customer behavior patterns. The recommendation system is rule-based rather than dynamically learned from user interactions, which limits adaptability. Also, both case studies use static datasets and do not handle real-time data updates or live customer behavior.

Future Work:

In future work, the project can be extended by integrating real-time transactional data so that recommendations and customer segmentation can be updated dynamically. More advanced recommendation techniques such as collaborative filtering, content-based filtering, or hybrid recommendation systems can be implemented instead of relying only on association rules and fixed rules. Deep learning models or ensemble approaches can also be explored to improve prediction accuracy in Case Study 2. For customer segmentation, additional features such as browsing behavior, customer reviews, product ratings, and return history can be included for better clustering. The advertisement recommendation system can be improved using Natural Language Processing (NLP) techniques instead of simple string matching to better understand product descriptions and customer interests. A dashboard-based interface can also be developed for visualization and real-time business decision support.

How the system can be improved:

The system can be made better by improving both analytical depth and practical usability. Hyperparameter tuning and model optimization can improve the performance of the Random Forest classifier and customer clustering models. Better handling of class imbalance can reduce misclassification in minority product categories. Instead of manual rule-based offer generation, adaptive recommendation systems can be created using customer feedback and purchase history. Incorporating sentiment analysis from customer reviews can provide additional insight into product demand and satisfaction. Cloud integration with platforms like Strapi, Cloudinary, or live databases can help make the project more scalable and production-ready. Finally, combining both online and offline retail data into a unified recommendation engine would create a stronger omnichannel retail solution, providing more accurate personalization and better business outcomes.